

EXPERIMENT – 36

Implementation of Affine Caesar Cipher

CODE:

```
#include <stdio.h>
#include <string.h>

int gcd(int a, int b)
{
    while (b)
    {
        int t = b;
        b = a % b;
        a = t;
    }
    return a;
}

int main()
{
    char text[100];
    int a, b, i;

    printf("Enter plaintext (CAPITAL letters): ");
    scanf("%s", text);

    printf("Enter a and b: ");
    scanf("%d%d", &a, &b);
```

```

if (gcd(a, 26) != 1)
{
    printf("Invalid value of a!\n");
    printf("For one-to-one encryption, gcd(a,26) must be 1.\n");
    return 0;
}

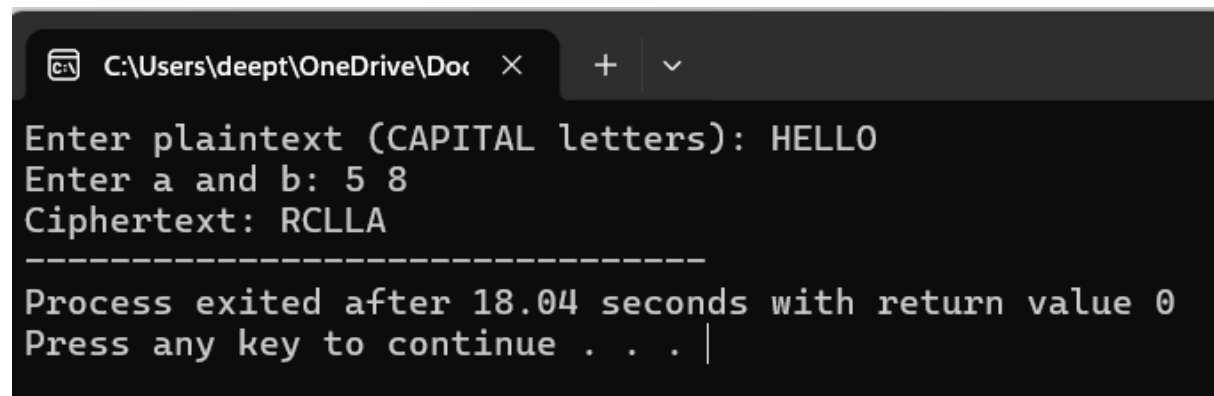
printf("Ciphertext: ");

for (i = 0; text[i] != '\0'; i++)
{
    int p = text[i] - 'A';
    int c = (a * p + b) % 26;
    printf("%c", c + 'A');
}

return 0;
}

```

OUTPUT:



```

C:\Users\deepthi\OneDrive\Documents
Enter plaintext (CAPITAL letters): HELLO
Enter a and b: 5 8
Ciphertext: RCLLA
-----
Process exited after 18.04 seconds with return value 0
Press any key to continue . . . |

```